

Unit 8. Generalised Linear Models II: Counts and the Exponential Family

1 Poisson Regression

FREES §12.1.

1.1 The Poisson distribution and exposure

Definition 8.1 (Poisson). y has the **Poisson** distribution with mean $\mu > 0$, written $y \sim \text{Pois}(\mu)$, if

$$\mathbf{P}\{y = j\} = \frac{\mu^j}{j!} e^{-\mu}, \quad j = 0, 1, 2, \dots \quad (1)$$

Proposition 8.2 (Moments and additivity). Let $y \sim \text{Pois}(\mu)$.

- (i) The probability generating function is $G(s) = \mathbf{E}[s^y] = e^{\mu(s-1)}$ for every $s \in \mathbb{R}$.
- (ii) $\mathbf{E}[y] = \mu$ and $\text{var}(y) = \mu$: the mean and the variance are equal, a property called **equidispersion**.
- (iii) If $y_1, \dots, y_m$ are independent with $y_j \sim \text{Pois}(\mu_j)$ then $\sum_j y_j \sim \text{Pois}(\sum_j \mu_j)$.

Proof. (i) $\mathbf{E}[s^y] = e^{-\mu} \sum_{j \geq 0} (\mu s)^j / j! = e^{-\mu} e^{\mu s}$, the series converging absolutely for every s .
(ii) Differentiating the series term by term, which is legitimate inside its interval of convergence, $G'(1) = \mathbf{E}[y] = \mu$ and $G''(1) = \mathbf{E}[y(y-1)] = \mu^2$, so $\text{var}(y) = \mu^2 + \mu - \mu^2 = \mu$.
(iii) The generating function of the sum is the product $\prod_j e^{\mu_j(s-1)} = e^{(\sum_j \mu_j)(s-1)}$ by independence. Two distributions on the non-negative integers with the same generating function are equal, since the coefficients of a convergent power series are determined by the function; and the right side is the generating function of $\text{Pois}(\sum_j \mu_j)$. $\square$

Definition 8.3 (Exposure). An **exposure** $E_i > 0$ is a known measure of the amount of risk observed: policy-years, vehicle-months, population at risk. The model with exposure is $\mathbf{E}[y_i] = E_i \mu_i$, where μ_i is a rate per unit of exposure.

Remark. Proposition 8.2(iii) is why exposures are natural for counts: a policy observed for two years, with an independent $\text{Pois}(\mu)$ count in each, produces a $\text{Pois}(2\mu)$ total. Fractional exposures extend the same relation by definition rather than by proof.

1.2 The regression model

Definition 8.4 (Poisson regression). Given non-random $\mathbf{x}_i \in \mathbb{R}^p$ with first coordinate 1 and exposures E_i , the **Poisson regression model** with **logarithmic link** takes $y_1, \dots, y_n$ independent with

$$y_i \sim \text{Pois}(\mu_i), \quad \ln \mu_i = \ln E_i + \mathbf{x}_i' \boldsymbol{\beta}. \quad (2)$$

The term $\ln E_i$, a covariate whose coefficient is fixed at 1, is called an **offset**.

Proposition 8.5 (Interpretation). Under (2):

- (i) $\mu_i = E_i \exp(\mathbf{x}_i' \boldsymbol{\beta})$, which is positive for every $\boldsymbol{\beta}$ and every $\mathbf{x}_i$;
- (ii) a unit increase in x_{ij} , the other coordinates and the exposure held fixed, multiplies the mean by e^{β_j} ;
- (iii) $\partial \ln \mu_i / \partial x_{ij} = \beta_j$, so β_j is the proportional change in the mean per unit change in the covariate;
- (iv) with a categorical design and a logarithmic link the fitted mean factorises, $\mu = B \cdot A_1 \cdot A_2 \cdots$, one factor per level: a **multiplicative** rating structure.

Proof. (i) Exponentiate (2). (ii) The ratio of the two means is $\exp\{(x_{ij} + 1)\beta_j - x_{ij}\beta_j\} = e^{\beta_j}$, every other term cancelling. (iii) Differentiate $\ln \mu_i = \ln E_i + \mathbf{x}'_i \boldsymbol{\beta}$. (iv) With $\mathbf{x}'_i \boldsymbol{\beta} = \beta_0 + \alpha_{a(i)} + \tau_{u(i)}$ for level indicators, $\mu_i = e^{\beta_0} e^{\alpha_{a(i)}} e^{\tau_{u(i)}}$. $\square$

Remark. The offset is not an estimated coefficient. Putting $\ln E_i$ in as an ordinary covariate estimates a coefficient γ and fits $\mu_i = E_i^\gamma e^{\mathbf{x}'_i \boldsymbol{\beta}}$, a different model; the offset is the restriction $\gamma = 1$, which is what proportionality of the mean to exposure asserts.

1.3 The likelihood

Theorem 8.6 (Score, information, concavity). For the model (2) the log-likelihood, score and information are

$$\ell(\boldsymbol{\beta}) = \sum_{i=1}^n \{y_i(\ln E_i + \mathbf{x}'_i \boldsymbol{\beta}) - E_i e^{\mathbf{x}'_i \boldsymbol{\beta}} - \ln y_i!\}, \quad (3)$$

$$\nabla \ell(\boldsymbol{\beta}) = \sum_{i=1}^n \mathbf{x}_i(y_i - \mu_i) = \mathbf{X}'(\mathbf{Y} - \boldsymbol{\mu}), \quad \mathcal{I}(\boldsymbol{\beta}) = \sum_{i=1}^n \mu_i \mathbf{x}_i \mathbf{x}'_i = \mathbf{X}' \mathbf{W} \mathbf{X}, \quad (4)$$

with $\mathbf{W} = \text{diag}(\mu_i)$. The Hessian is $-\mathcal{I}(\boldsymbol{\beta})$ identically, so observed and expected information agree; ℓ is concave, and strictly concave when $\text{rank}(\mathbf{X}) = p$.

Proof. Summing the logarithm of (1) at the observed value of each y_i , and using $\ln \mu_i = \ln E_i + \mathbf{x}_i' \boldsymbol{\beta}$ gives (3). Differentiating, $\partial \mu_i / \partial \boldsymbol{\beta} = \mu_i \mathbf{x}_i$, so $\nabla \ell = \sum_i \mathbf{x}_i y_i - \sum_i \mu_i \mathbf{x}_i$, which is (4), and differentiating once more gives $\nabla^2 \ell = -\sum_i \mu_i \mathbf{x}_i \mathbf{x}_i'$, free of the y_i and hence equal to its own expectation. That matrix is negative semi-definite because every $\mu_i > 0$, and negative definite exactly when the $\mathbf{x}_i$ span $\mathbb{R}^p$. $\square$

Corollary 8.7 (What the score equations force). At the maximum likelihood estimate, $\sum_i x_{ij} \hat{\mu}_i = \sum_i x_{ij} y_i$ for every covariate j . With an intercept, $\sum_i \hat{\mu}_i = \sum_i y_i$: the fitted counts reproduce the observed total, and within each level of a categorical covariate.

Proof. Setting (4) to zero is $\mathbf{X}'(\mathbf{Y} - \hat{\boldsymbol{\mu}}) = \mathbf{0}$; read coordinate by coordinate, with $x_{i0} \equiv 1$ for the intercept and x_{ij} the indicator of a level. $\square$

Theorem 8.8 (Iteratively reweighted least squares). With $\eta_i = \ln E_i + \mathbf{x}_i' \boldsymbol{\beta}^{(m)}$, $\mu_i = e^{\eta_i}$, $w_i = \mu_i$ and working response $z_i = \mathbf{x}_i' \boldsymbol{\beta}^{(m)} + (y_i - \mu_i)/\mu_i$, the Newton step is $\boldsymbol{\beta}^{(m+1)} = (\mathbf{X}' \mathbf{W} \mathbf{X})^{-1} \mathbf{X}' \mathbf{W} \mathbf{z}$.

Proof. The Newton step is $\boldsymbol{\beta}^{(m)} + (\mathbf{X}' \mathbf{W} \mathbf{X})^{-1} \mathbf{X}' (\mathbf{Y} - \boldsymbol{\mu})$ by Theorem 8.6. Since $\mathbf{W} \mathbf{z} = \mathbf{W} \mathbf{X} \boldsymbol{\beta}^{(m)} + (\mathbf{Y} - \boldsymbol{\mu})$ by the definition of z_i , the stated formula equals $(\mathbf{X}' \mathbf{W} \mathbf{X})^{-1} \{ \mathbf{X}' \mathbf{W} \mathbf{X} \boldsymbol{\beta}^{(m)} + \mathbf{X}' (\mathbf{Y} - \boldsymbol{\mu}) \}$, which is the same vector. $\square$

1.4 Asymptotics

Theorem 8.9 (Sampling distribution). Consider (2) with true parameter $\boldsymbol{\beta}_0$ and suppose

(D1) $|\mathbf{x}_i| \leq M$ and $c \leq E_i \leq C$ for constants free of i ;

(D2) $\mathbf{F}_n = n^{-1} \sum_{i \leq n} \mu_i \mathbf{x}_i \mathbf{x}_i' \rightarrow \mathbf{F}$ positive definite, the μ_i evaluated at $\boldsymbol{\beta}_0$.

Then with probability tending to one $\hat{\boldsymbol{\beta}}$ exists and is unique, $\hat{\boldsymbol{\beta}} \xrightarrow{p} \boldsymbol{\beta}_0$, and

$$\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{F}^{-1}), \quad n^{-1} \mathcal{I}(\hat{\boldsymbol{\beta}}) \xrightarrow{p} \mathbf{F}. \quad (5)$$

The Wald and likelihood ratio statistics of Unit 7 are χ_q^2 in the limit.

Proof. The three hypotheses of the concave maximum likelihood theorem of the prerequisites are checked with $\Theta = \mathbb{R}^p$ and ℓ_i the i -th summand of (3).

Concavity and smoothness. Theorem 8.6.

The score condition. $\nabla \ell_i(\beta_0) = \mathbf{x}_i(y_i - \mu_i)$ has mean zero and variance $\mu_i \mathbf{x}_i \mathbf{x}_i'$, so the average variance is $\mathbf{F}_n \rightarrow \mathbf{F}$ by (D2). The summands are **not** bounded, since y_i is not, so the bounded-summand corollary of the prerequisites does not apply; the Lyapunov corollary does. Under (D1) the means $\mu_i = E_i e^{\mathbf{x}_i' \beta_0}$ lie in a fixed interval $[m_-, m_+]$ with $m_- > 0$, and the fourth central moment of a Poisson variable is $\mu + 3\mu^2$, so

$$E |\mathbf{x}_i(y_i - \mu_i)|^4 = |\mathbf{x}_i|^4 E(y_i - \mu_i)^4 \leq M^4(m_+ + 3m_+^2),$$

bounded uniformly in i . Taking $\delta = 2$ in the Lyapunov corollary gives $n^{-1/2} \mathbf{U}_n \xrightarrow{d} N(\mathbf{0}, \mathbf{F})$.

The curvature condition. $\mathbf{H}_n(\beta) = \sum_i E_i e^{\mathbf{x}_i' \beta} \mathbf{x}_i \mathbf{x}_i'$ is non-random and equals $n\mathbf{F}_n$ at β_0 , so $n^{-1} \mathbf{H}_n(\beta_0) \rightarrow \mathbf{F}$: the limiting curvature is the same $\mathbf{F}$ as the limiting score variance, which is the information equality for this correctly specified model. For the Lipschitz bound, on $|\beta - \beta_0| \leq \delta_0$ the exponents $\mathbf{x}_i' \beta$ lie in a fixed compact interval, so $|e^{\mathbf{x}_i' \beta} - e^{\mathbf{x}_i' \beta_0}| \leq K |\mathbf{x}_i| |\beta - \beta_0|$ by the mean

value theorem with K the largest value of the exponential on that interval, and hence

$$\|\mathbf{H}_n(\boldsymbol{\beta}) - \mathbf{H}_n(\boldsymbol{\beta}_0)\| \leq \sum_{i=1}^n CKM^3 |\boldsymbol{\beta} - \boldsymbol{\beta}_0| = nCKM^3 |\boldsymbol{\beta} - \boldsymbol{\beta}_0|.$$

The conclusions are the existence, consistency, asymptotic normality, information and likelihood ratio statements of that theorem and its corollaries. $\square$

Remark. The fourth moment of a centred Poisson variable is used only to make the Lyapunov ratio finite; any $\delta > 0$ would do. Condition (D1) keeps the means away from 0 and ∞ , which makes the moment bound uniform: a design in which some μ_i drifts to zero has no such bound, and the asymptotics of a cell with almost no exposure are not covered.

2 Overdispersion and Other Count Models

FREES §12.3–12.4.

2.1 Detecting overdispersion

Definition 8.10 (Over- and underdispersion). Counts with $\text{var}(y_i) > \mathbf{E}[y_i]$ are **overdispersed** relative to the Poisson, and **underdispersed** when the inequality is reversed.

Proposition 8.11 (A moment test). Let $T_i = (y_i - \mu_i)^2 - y_i$ with $\mu_i = \mathbf{E}[y_i]$. If y_i is Poisson then $\mathbf{E}[T_i] = 0$. If instead $\text{var}(y_i) = \mu_i + \alpha v(\mu_i)$ for some α and a known function $v > 0$, then $\mathbf{E}[T_i] = \alpha v(\mu_i)$, so the least squares slope of T_i on $v(\mu_i)$ through the origin estimates α .

Proof. $\mathbf{E}[(y_i - \mu_i)^2] = \text{var}(y_i)$ and $\mathbf{E}[y_i] = \mu_i$, so $\mathbf{E}[T_i] = \text{var}(y_i) - \mu_i$, which is 0 under equidispersion and $\alpha v(\mu_i)$ under the stated alternative. Regressing T_i on $v(\mu_i)$ without an intercept has expectation α for its slope, the design being non-random once the μ_i are. $\square$

Definition 8.12 (Pearson statistic and dispersion estimate). With fitted means $\hat{\mu}_i$ and p estimated coefficients,

$$X^2 = \sum_{i=1}^n \frac{(y_i - \hat{\mu}_i)^2}{\hat{\mu}_i}, \quad \hat{\phi} = \frac{X^2}{n - p}. \quad (6)$$

Proposition 8.13 (Consistency of $\hat{\phi}$). Suppose $\mathbf{E}[y_i] = \mu_i$ and $\text{var}(y_i) = \phi\mu_i$, that $\mu_i \in [m_-, m_+]$ with $m_- > 0$, that $\sup_i \mathbf{E}[(y_i - \mu_i)^4] < \infty$, and that $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0 = O_p(n^{-1/2})$ with $|\mathbf{x}_i| \leq M$. Then $\hat{\phi} \xrightarrow{p} \phi$.

Proof. Write $r_i = y_i - \mu_i$ and $\delta_i = \hat{\mu}_i - \mu_i$. Since $\mu(\eta) = E_i e^\eta$ is Lipschitz on the compact range of $\mathbf{x}_i' \boldsymbol{\beta}$ for $\boldsymbol{\beta}$ near $\boldsymbol{\beta}_0$, $\max_i |\delta_i| \leq K |\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0| = O_p(n^{-1/2})$.

First, the infeasible version. $\mathbf{E}[r_i^2/\mu_i] = \phi$ and $\text{var}(r_i^2/\mu_i) \leq \mathbf{E}[r_i^4]/m_-^2$, bounded, so by Chebyshev

$$\tilde{\phi} = \frac{1}{n} \sum_{i=1}^n \frac{r_i^2}{\mu_i} \xrightarrow{p} \phi. \quad (7)$$

Next, the difference. On the event $\max_i |\delta_i| \leq m_-/2$, whose probability tends to 1, every $\hat{\mu}_i \geq m_-/2$ and

$$\frac{(r_i - \delta_i)^2}{\hat{\mu}_i} - \frac{r_i^2}{\mu_i} = -\frac{r_i^2 \delta_i}{\mu_i \hat{\mu}_i} + \frac{\delta_i^2 - 2r_i \delta_i}{\hat{\mu}_i}.$$

Averaging over i and bounding, with $\Delta = \max_i |\delta_i|$,

$$\left| \frac{1}{n} \sum_i \left\{ \frac{(r_i - \delta_i)^2}{\hat{\mu}_i} - \frac{r_i^2}{\mu_i} \right\} \right| \leq \frac{2\Delta}{m_-^2} \cdot \frac{1}{n} \sum_i r_i^2 + \frac{2}{m_-} \left(\Delta^2 + 2\Delta \cdot \frac{1}{n} \sum_i |r_i| \right).$$

Both averages $n^{-1} \sum_i r_i^2$ and $n^{-1} \sum_i |r_i|$ are $O_p(1)$, their expectations being bounded, and $\Delta = O_p(n^{-1/2})$, so the whole bound is $o_p(1)$. With $n/(n-p) \rightarrow 1$ and (7), $\hat{\phi} \xrightarrow{p} \phi$. $\square$

Remark. The residual deviance divided by $n-p$ is a second dispersion estimate. The two are not equal and need not be close: the Pearson statistic weights by $\hat{\mu}_i$ and the deviance by a logarithmic discrepancy.

2.2 Quasi-Poisson and the sandwich

Theorem 8.14 (Quasi-Poisson). Suppose $y_1, \dots, y_n$ are independent with

$$\mathbb{E}[y_i] = \mu_i = E_i e^{\mathbf{x}_i' \beta_0}, \quad \text{var}(y_i) = \phi \mu_i, \quad (8)$$

and that $\sup_i \mathbb{E} |y_i - \mu_i|^{2+\delta} < \infty$ for some $\delta > 0$, with (D1) and (D2) of Theorem 8.9. Let $\hat{\beta}$

maximise the Poisson log-likelihood (3), whatever the true distribution. Then

- (i) $\hat{\boldsymbol{\beta}}$ is the same vector as under the Poisson model: the estimator depends on the data only through the score equations $\mathbf{X}'(\mathbf{Y} - \boldsymbol{\mu}) = \mathbf{0}$;
- (ii) $\hat{\boldsymbol{\beta}} \xrightarrow{p} \boldsymbol{\beta}_0$ and $\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) \xrightarrow{d} N(\mathbf{0}, \phi \mathbf{F}^{-1})$;
- (iii) the correct standard errors are $\sqrt{\phi}$ times the Poisson ones.

Proof. (i) is the definition of $\hat{\boldsymbol{\beta}}$: nothing outside (3) enters its computation.

(ii) Apply the concave maximum likelihood theorem of the prerequisites with ℓ_n the Poisson log-likelihood, which is a criterion function rather than the true log-likelihood. The hypotheses are checked as in Theorem 8.9, with two changes. The score still has mean zero, $\mathbf{E}[\mathbf{x}_i(y_i - \mu_i)] = \mathbf{0}$, because the **mean** is correctly specified; that is all the theorem requires. Its variance is now $\phi \mu_i \mathbf{x}_i \mathbf{x}_i'$, so $\mathbf{F}_n \rightarrow \phi \mathbf{F}$, and the Lyapunov corollary of the prerequisites gives the central limit theorem under the assumed $(2 + \delta)$ -moment bound. The curvature is unchanged, $n^{-1} \mathbf{H}_n(\boldsymbol{\beta}_0) = n^{-1} \sum_i \mu_i \mathbf{x}_i \mathbf{x}_i' \rightarrow \mathbf{F}$, so the limiting curvature is $\mathbf{G} = \mathbf{F}$ while the limiting score variance is $\phi \mathbf{F}$.

The sandwich of that theorem is

$$\mathbf{G}^{-1}(\phi \mathbf{F}) \mathbf{G}^{-1} = \mathbf{F}^{-1}(\phi \mathbf{F}) \mathbf{F}^{-1} = \phi \mathbf{F}^{-1}.$$

(iii) Standard errors are square roots of diagonal entries of n^{-1} times the asymptotic variance, so multiplying that variance by ϕ multiplies each by $\sqrt{\phi}$. $\square$

Remark. Under (8) the information equality fails by exactly the factor ϕ , and the whole effect of overdispersion on inference is that constant. The point estimates need no correction, the standard errors need one factor, and the likelihood ratio statistic needs dividing by $\hat{\phi}$ before being referred to χ^2 , since it is a difference of criterion values, not of log-likelihoods.

Proposition 8.15 (Fully robust standard errors). If only $\mathbf{E}[y_i] = \mu_i$ is assumed, with $\text{var}(y_i) = \sigma_i^2$ unrestricted and the moment conditions of Theorem 8.14, then $\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{G}^{-1} \mathbf{F} \mathbf{G}^{-1})$

with $\mathbf{G} = \lim n^{-1} \sum_i \mu_i \mathbf{x}_i \mathbf{x}_i'$ and $\mathbf{F} = \lim n^{-1} \sum_i \sigma_i^2 \mathbf{x}_i \mathbf{x}_i'$, estimated by

$$\widehat{\text{var}}(\hat{\boldsymbol{\beta}}) = \left(\sum_i \hat{\mu}_i \mathbf{x}_i \mathbf{x}_i' \right)^{-1} \left(\sum_i (y_i - \hat{\mu}_i)^2 \mathbf{x}_i \mathbf{x}_i' \right) \left(\sum_i \hat{\mu}_i \mathbf{x}_i \mathbf{x}_i' \right)^{-1}. \quad (9)$$

Proof. The argument of Theorem 8.14(ii) is unchanged except that the score variance is $\sum_i \sigma_i^2 \mathbf{x}_i \mathbf{x}_i'$, giving $\mathbf{F}$ as stated and the sandwich of the prerequisites' theorem. The middle matrix of (9) is consistent for $n\mathbf{F}$ because $\mathbf{E}[(y_i - \mu_i)^2] = \sigma_i^2$ and the plug-in error is $o_p(1)$ by the argument of Proposition 8.13. $\square$

Remark. Quasi-Poisson forces the ratio of the robust to the model-based standard error to be the same constant $\sqrt{\hat{\phi}}$ for every coefficient. The sandwich does not, and a wide spread of those ratios is evidence that $\text{var}(y_i) = \phi\mu_i$ is itself wrong (§4.3).

2.3 The negative binomial

Definition 8.16 (Negative binomial). For $\mu > 0$ and $\theta > 0$, $y \sim \text{NB}(\mu, \theta)$ if

$$\mathbf{P}\{y = j\} = \frac{\Gamma(j + \theta)}{j! \Gamma(\theta)} \left(\frac{\theta}{\theta + \mu}\right)^\theta \left(\frac{\mu}{\theta + \mu}\right)^j, \quad j = 0, 1, 2, \dots \quad (10)$$

Theorem 8.17 (Poisson–gamma mixture). Let λ have the gamma density with shape θ and rate θ/μ , so that $\mathbf{E}[\lambda] = \mu$ and $\text{var}(\lambda) = \mu^2/\theta$, and let $y \mid \lambda \sim \text{Pois}(\lambda)$. Then $y \sim \text{NB}(\mu, \theta)$, and

$$\mathbf{E}[y] = \mu, \quad \text{var}(y) = \mu + \frac{\mu^2}{\theta}. \quad (11)$$

Proof. Write $\rho = \theta/\mu$. Integrating out λ ,

$$\mathbf{P}\{y = j\} = \int_0^\infty \frac{\lambda^j e^{-\lambda}}{j!} \frac{\rho^\theta}{\Gamma(\theta)} \lambda^{\theta-1} e^{-\rho\lambda} d\lambda = \frac{\rho^\theta}{j! \Gamma(\theta)} \int_0^\infty \lambda^{j+\theta-1} e^{-(1+\rho)\lambda} d\lambda.$$

The integral is $\Gamma(j + \theta)/(1 + \rho)^{j+\theta}$, by the definition of the gamma function after the substitution

$u = (1 + \rho)\lambda$, so

$$\mathbf{P}\{y = j\} = \frac{\Gamma(j + \theta)}{j! \Gamma(\theta)} \left(\frac{\rho}{1 + \rho}\right)^\theta \left(\frac{1}{1 + \rho}\right)^j,$$

and $\rho/(1 + \rho) = \theta/(\theta + \mu)$ while $1/(1 + \rho) = \mu/(\theta + \mu)$, which is (10).

For the moments, the tower property of the prerequisites gives $\mathbf{E}[y] = \mathbf{E}[\lambda] = \mu$, and the conditional variance decomposition there gives $\text{var}(y) = \mathbf{E}[\text{var}(y \mid \lambda)] + \text{var}(\mathbf{E}[y \mid \lambda]) = \mathbf{E}[\lambda] + \text{var}(\lambda) = \mu + \mu^2/\theta$. $\square$

Corollary 8.18 (The Poisson is the limit). For fixed j and μ , $\mathbf{P}\{y = j\} \rightarrow \mu^j e^{-\mu}/j!$ as $\theta \rightarrow \infty$, and $\text{var}(y) \downarrow \mu$.

Proof. The variance statement is immediate from (11). For the mass function,

$$\frac{\Gamma(j + \theta)}{\Gamma(\theta)\theta^j} = \prod_{l=0}^{j-1} \frac{\theta + l}{\theta} \longrightarrow 1,$$

a product of j factors each tending to 1; and rewriting (10) as

$$\frac{\Gamma(j + \theta)}{\Gamma(\theta)\theta^j} \cdot \frac{\mu^j}{j!} \cdot \left(1 + \frac{\mu}{\theta}\right)^{-\theta} \left(1 + \frac{\mu}{\theta}\right)^{-j},$$

the third factor tends to $e^{-\mu}$ and the fourth to 1. □

Proposition 8.19 (Concavity in $\boldsymbol{\beta}$ for fixed θ). With $\mu_i = E_i e^{\mathbf{x}_i' \boldsymbol{\beta}}$ and θ fixed, the negative binomial log-likelihood is concave in $\boldsymbol{\beta}$, strictly so when $\text{rank}(\mathbf{X}) = p$.

Proof. Dropping terms free of $\boldsymbol{\beta}$, the i -th contribution is $y_i \ln \mu_i - (y_i + \theta) \ln(\theta + \mu_i)$. As a function of $\eta = \mathbf{x}_i' \boldsymbol{\beta}$, with $\mu = E_i e^\eta$ and $d\mu/d\eta = \mu$, the first derivative is

$$y_i - \frac{(y_i + \theta)\mu}{\theta + \mu}, \quad \text{and the second is} \quad -\frac{(y_i + \theta)\theta\mu}{(\theta + \mu)^2} \leq 0,$$

using $d/d\eta\{\mu/(\theta + \mu)\} = \theta\mu/(\theta + \mu)^2$. Composing with the linear map $\boldsymbol{\beta} \mapsto \mathbf{x}_i' \boldsymbol{\beta}$ preserves concavity, and the second derivative is strictly negative unless $y_i = 0$ and $\theta = 0$, so the sum is strictly concave when the $\mathbf{x}_i$ span. □

Remark. For fixed θ the model is a linear exponential family in μ with variance function $V(\mu) = \mu + \mu^2/\theta$ (§3). With θ estimated it is not, and the joint likelihood in $(\boldsymbol{\beta}, \theta)$ is maximised by alternating between the concave problem of Proposition 8.19 and a one-dimensional search in θ . The Poisson sits on the boundary $\theta = \infty$ of the parameter space, so a likelihood ratio test of Poisson against negative binomial does not have the χ_1^2 limit of Unit 7; the comparison is made by an information criterion.

2.4 Excess zeros

Definition 8.20 (Zero-inflated and hurdle models). Let g_i be a count mass function with mean μ_i , and $\pi_i \in (0, 1)$.

(i) The **zero-inflated** model mixes a point mass at 0 with g_i :

$$\mathbf{P}\{y_i = j\} = \begin{cases} \pi_i + (1 - \pi_i)g_i(0), & j = 0, \\ (1 - \pi_i)g_i(j), & j \geq 1. \end{cases} \quad (12)$$

(ii) The **hurdle** model replaces the zero probability outright:

$$P\{y_i = j\} = \begin{cases} \pi_i, & j = 0, \\ k_i g_i(j), & j \geq 1, \end{cases} \quad k_i = \frac{1 - \pi_i}{1 - g_i(0)}. \quad (13)$$

In both, π_i is modelled by a binary regression, usually logistic, and μ_i by a log link.

Theorem 8.21 (Moments). Let g_i be the $\text{Pois}(\mu_i)$ mass function.

(i) Under (12), $E[y_i] = (1 - \pi_i)\mu_i$ and

$$\text{var}(y_i) = (1 - \pi_i)\mu_i + \pi_i(1 - \pi_i)\mu_i^2 = E[y_i] + \pi_i(1 - \pi_i)\mu_i^2, \quad (14)$$

so a zero-inflated Poisson is always overdispersed.

(ii) Under (13), $E[y_i] = k_i\mu_i$ and $\text{var}(y_i) = k_i\mu_i + k_i(1 - k_i)\mu_i^2$, which exceeds the mean when $k_i < 1$ and falls below it when $k_i > 1$.

Proof. (i) Let B be the indicator of the point mass, so $\mathbf{P}\{B = 1\} = \pi_i$ and $y_i = (1 - B)N$ with $N \sim \text{Pois}(\mu_i)$ independent of B . Then $\mathbf{E}[y_i] = (1 - \pi_i)\mu_i$ and, since $(1 - B)^2 = 1 - B$,

$$\mathbf{E}[y_i^2] = \mathbf{E}[(1 - B)N^2] = (1 - \pi_i)(\mu_i + \mu_i^2).$$

Subtracting the square of the mean, $\text{var}(y_i) = (1 - \pi_i)\mu_i + (1 - \pi_i)\mu_i^2 - (1 - \pi_i)^2\mu_i^2$, and the last two terms combine to $(1 - \pi_i)\mu_i^2\{1 - (1 - \pi_i)\} = \pi_i(1 - \pi_i)\mu_i^2$. That quantity is strictly positive for $\pi_i \in (0, 1)$, so the variance exceeds the mean.

(ii) Under (13), $\mathbf{E}[y_i] = k_i \sum_{j \geq 1} j g_i(j) = k_i \mu_i$ and $\mathbf{E}[y_i^2] = k_i \sum_{j \geq 1} j^2 g_i(j) = k_i(\mu_i + \mu_i^2)$, the $j = 0$ terms contributing nothing in either sum. Subtracting gives $k_i \mu_i + k_i \mu_i^2 - k_i^2 \mu_i^2$. $\square$

Theorem 8.22 (The hurdle likelihood separates). Let $\pi_i = \pi(\mathbf{x}_i; \boldsymbol{\gamma})$ and $g_i = g(\cdot; \mathbf{x}_i, \boldsymbol{\beta})$ with $\boldsymbol{\gamma}$ and $\boldsymbol{\beta}$ functionally unrelated. Then the log-likelihood of (13) is

$$\ell(\boldsymbol{\gamma}, \boldsymbol{\beta}) = \underbrace{\sum_i \left[\mathbf{1}\{y_i = 0\} \ln \pi_i + \mathbf{1}\{y_i > 0\} \ln(1 - \pi_i) \right]}_{\ell_1(\boldsymbol{\gamma})} + \underbrace{\sum_{i: y_i > 0} \ln \frac{g_i(y_i)}{1 - g_i(0)}}_{\ell_2(\boldsymbol{\beta})}, \quad (15)$$

so the two halves may be maximised separately: ℓ_1 is a binary regression on $\mathbb{1}\{y_i = 0\}$ and ℓ_2 is a zero-truncated count regression.

Proof. Summing the logarithm of (13) at each observed value, an observation with $y_i = 0$ contributes $\ln \pi_i$ and one with $y_i > 0$ contributes $\ln k_i + \ln g_i(y_i) = \ln(1 - \pi_i) - \ln(1 - g_i(0)) + \ln g_i(y_i)$. Collecting the terms in π_i and the terms in g_i gives (15), and since no parameter appears in both sums, maximising the total is maximising each. $\square$

Corollary 8.23 (The hurdle reproduces the zero count). If the zero part is a logistic regression with an intercept only, then $\hat{\pi} = n^{-1} \#\{i : y_i = 0\}$ and the fitted expected number of zeros equals the observed number exactly.

Proof. By Theorem 8.22 the zero part is a Bernoulli log-likelihood for $\mathbb{1}\{y_i = 0\}$ with a single parameter, and by the score identity of Unit 7 the fitted probability reproduces the observed proportion. The fitted expected count of zeros is $n\hat{\pi}$, which is that proportion times n . $\square$

Remark. The zero-inflated model has no such property: its zero probability $\pi_i + (1 - \pi_i)e^{-\mu_i}$ mixes both parts, and the two parameters are estimated jointly from the whole sample. The two models also mean different things. Zero inflation says some subjects can only produce a zero; the hurdle says every subject faces one mechanism for crossing zero and another for how far past it to go.

3 The Linear Exponential Family

FREES §13.1–13.3, §13.5–13.6, §13.9.

3.1 Definition and moments

Definition 8.24 (Linear exponential family). Fix $\phi > 0$ and a σ -finite measure ν on $\mathbb{R}$ that is not concentrated at a single point. Put

$$\Theta = \left\{ \theta \in \mathbb{R} : \int e^{y\theta/\phi} d\nu(y) < \infty \right\},$$

assume its interior Θ° is non-empty, and define the **cumulant function**

$$b(\theta) = \phi \ln \int e^{y\theta/\phi} d\nu(y), \quad \theta \in \Theta^\circ. \quad (16)$$

The **linear exponential family** generated by (ν, ϕ) is the set of densities with respect to ν

$$f(y; \theta, \phi) = \exp\left\{\frac{y\theta - b(\theta)}{\phi}\right\}, \quad \theta \in \Theta^\circ, \quad (17)$$

with θ the **natural parameter** and ϕ the **dispersion**.

Remark. Writing $d\nu(y) = e^{S(y, \phi)} d\lambda(y)$ for a reference measure λ recovers the form $\exp\{(y\theta - b(\theta))/\phi + S(y, \phi)\}$ in which the family is usually displayed. The definition above makes $\int f d\nu = 1$ true by construction rather than by assumption.

Theorem 8.25 (Moments). On Θ° the function b is twice differentiable and, for y distributed

according to $f(\cdot; \theta, \phi)$,

$$\mathbb{E}_\theta[y] = b'(\theta), \quad \text{var}_\theta(y) = \phi b''(\theta). \quad (18)$$

Consequently b is strictly convex on Θ° and b' is strictly increasing.

Proof. Write $A(\theta) = \int e^{y\theta/\phi} d\nu(y)$, so that $b = \phi \ln A$. Fix $\theta_0 \in \Theta^\circ$ and $\delta > 0$ with $[\theta_0 - 2\delta, \theta_0 + 2\delta] \subset \Theta$.

Domination. For $k \geq 0$ and $u \geq 0$ the inequality $e^u \geq u^k/k!$ gives $|y|^k \leq k!(\phi/\delta)^k e^{|y|\delta/\phi}$. For $|\theta - \theta_0| < \delta$,

$$e^{|y|\delta/\phi} e^{y\theta/\phi} = \begin{cases} e^{y(\theta+\delta)/\phi} \leq e^{y(\theta_0+2\delta)/\phi}, & y \geq 0, \\ e^{y(\theta-\delta)/\phi} \leq e^{y(\theta_0-2\delta)/\phi}, & y < 0, \end{cases}$$

in each case because the exponent is increased. Hence

$$|y|^k e^{y\theta/\phi} \leq k! (\phi/\delta)^k \{e^{y(\theta_0+2\delta)/\phi} + e^{y(\theta_0-2\delta)/\phi}\} =: g_k(y), \quad (19)$$

and g_k is ν -integrable because $\theta_0 \pm 2\delta \in \Theta$.

Differentiation. Applying the differentiation-under-the-integral theorem of the prerequisites

twice, with g_1 and g_2 of (19) as dominating functions, A is twice differentiable on $(\theta_0 - \delta, \theta_0 + \delta)$ with

$$A'(\theta) = \frac{1}{\phi} \int y e^{y\theta/\phi} d\nu(y), \quad A''(\theta) = \frac{1}{\phi^2} \int y^2 e^{y\theta/\phi} d\nu(y).$$

Since $f(y; \theta, \phi) = e^{y\theta/\phi}/A(\theta)$ by (16) and (17), the two integrals are $A(\theta) \mathbf{E}_\theta[y]$ and $A(\theta) \mathbf{E}_\theta[y^2]$, so $A' = A \mathbf{E}_\theta[y]/\phi$ and $A'' = A \mathbf{E}_\theta[y^2]/\phi^2$. Therefore

$$b'(\theta) = \phi \frac{A'}{A} = \mathbf{E}_\theta[y], \quad b''(\theta) = \phi \frac{A''A - (A')^2}{A^2} = \frac{1}{\phi} \{ \mathbf{E}_\theta[y^2] - (\mathbf{E}_\theta[y])^2 \} = \frac{\text{var}_\theta(y)}{\phi}.$$

Finally $\text{var}_\theta(y) > 0$ because ν , and hence the law of y , is not concentrated at a point; so $b'' > 0$. $\square$

Definition 8.26 (Variance function; weights). By Theorem 8.25 the map $\theta \mapsto \mu = b'(\theta)$ is a strictly increasing bijection onto its range, so $\theta = (b')^{-1}(\mu)$ and

$$\text{var}(y) = \phi V(\mu), \quad V(\mu) = b''((b')^{-1}(\mu)), \quad (20)$$

the **variance function**. When observation i carries a known weight $w_i > 0$ its dispersion is

$$\phi_i = \phi/w_i, \text{ so } \text{var}(y_i) = \phi V(\mu_i)/w_i.$$

Family	$b(\theta)$	$\mu = b'(\theta)$	$V(\mu)$	canonical link $(b')^{-1}(\mu)$
Normal	$\theta^2/2$	θ	1	μ
Bernoulli	$\ln(1 + e^\theta)$	$e^\theta/(1 + e^\theta)$	$\mu(1 - \mu)$	$\text{logit } \mu$
Poisson	e^θ	e^θ	μ	$\ln \mu$
Negative binomial	$-\theta_* \ln(1 - e^\theta)$	$\theta_* e^\theta/(1 - e^\theta)$	$\mu + \mu^2/\theta_*$	$\ln\{\mu/(\mu + \theta_*)\}$
Gamma	$-\ln(-\theta)$	$-1/\theta$	μ^2	$-1/\mu$
Inverse Gaussian	$-\sqrt{-2\theta}$	$(-2\theta)^{-1/2}$	μ^3	$-1/(2\mu^2)$
Tweedie, $1 < p < 2$	see §3.4		μ^p	$\mu^{1-p}/(1 - p)$

The negative binomial row treats its shape θ_* as known, which makes it a one-parameter family; the Bernoulli and Poisson rows have $\phi = 1$.

3.2 Link functions

Definition 8.27 (Generalised linear model). A **generalised linear model** consists of independent $y_1, \dots, y_n$ from one linear exponential family with weights w_i , together with a strictly monotone differentiable **link** g and the systematic component

$$g(\mu_i) = \eta_i = \mathbf{x}_i' \boldsymbol{\beta}. \quad (21)$$

The link is **canonical** when $g = (b')^{-1}$, equivalently when $\theta_i = \eta_i$.

Proposition 8.28 (Why the canonical link is special). Under the canonical link the log-likelihood is $\ell(\boldsymbol{\beta}) = \phi^{-1} \sum_i w_i \{y_i \mathbf{x}_i' \boldsymbol{\beta} - b(\mathbf{x}_i' \boldsymbol{\beta})\} + \text{const}$, which is concave in $\boldsymbol{\beta}$, strictly so when $\text{rank}(\mathbf{X}) = p$; the observed and expected information coincide; and the score equations are $\sum_i w_i (y_i - \mu_i) \mathbf{x}_i = \mathbf{0}$.

Proof. Substituting $\theta_i = \mathbf{x}_i' \boldsymbol{\beta}$ into (17) with dispersion ϕ/w_i gives the displayed form. Each

$-b(\mathbf{x}'_i\boldsymbol{\beta})$ is concave because b is convex (Theorem 8.25) and $\boldsymbol{\beta} \mapsto \mathbf{x}'_i\boldsymbol{\beta}$ is linear. Differentiating,

$$\nabla \ell = \frac{1}{\phi} \sum_i w_i \{y_i - b'(\mathbf{x}'_i\boldsymbol{\beta})\} \mathbf{x}_i = \frac{1}{\phi} \sum_i w_i (y_i - \mu_i) \mathbf{x}_i, \quad -\nabla^2 \ell = \frac{1}{\phi} \sum_i w_i b''(\mathbf{x}'_i\boldsymbol{\beta}) \mathbf{x}_i \mathbf{x}'_i,$$

the second free of the y_i and hence equal to its expectation, and positive definite when the $\mathbf{x}_i$ span, since $b'' > 0$. $\square$

Theorem 8.29 (Score and information, general link). For any differentiable strictly monotone link,

$$\nabla \ell(\boldsymbol{\beta}) = \sum_{i=1}^n \frac{w_i}{\phi} \frac{y_i - \mu_i}{V(\mu_i)} \frac{d\mu_i}{d\eta_i} \mathbf{x}_i, \quad \mathcal{I}(\boldsymbol{\beta}) = \sum_{i=1}^n \frac{w_i}{\phi} \frac{1}{V(\mu_i)} \left(\frac{d\mu_i}{d\eta_i} \right)^2 \mathbf{x}_i \mathbf{x}'_i. \quad (22)$$

Under the canonical link $d\mu_i/d\eta_i = V(\mu_i)$ and (22) reduces to the expressions of Proposition 8.28.

Proof. With dispersion ϕ/w_i , the i -th log-likelihood term is $w_i\{y_i\theta_i - b(\theta_i)\}/\phi$ up to a constant, so $\partial \ell_i / \partial \theta_i = w_i(y_i - \mu_i)/\phi$. The chain rule through $\theta_i \mapsto \mu_i \mapsto \eta_i \mapsto \boldsymbol{\beta}$ needs $d\theta_i/d\mu_i$: differentiating $\mu_i = b'(\theta_i)$ gives $d\mu_i/d\theta_i = b''(\theta_i) = V(\mu_i)$, so $d\theta_i/d\mu_i = 1/V(\mu_i)$. Multiplying by $d\mu_i/d\eta_i$ and

$\partial\eta_i/\partial\boldsymbol{\beta} = \mathbf{x}_i$ gives the score.

The information is the variance of the score. The summands are independent with mean zero and $\text{var}(y_i) = \phi V(\mu_i)/w_i$, so

$$\mathcal{I}(\boldsymbol{\beta}) = \sum_i \frac{w_i^2}{\phi^2 V(\mu_i)^2} \left(\frac{d\mu_i}{d\eta_i} \right)^2 \text{var}(y_i) \mathbf{x}_i \mathbf{x}_i',$$

and substituting $\text{var}(y_i)$ leaves (22). For the canonical link $\eta_i = \theta_i$, so $d\mu_i/d\eta_i = b''(\theta_i) = V(\mu_i)$ and the ratio $(d\mu_i/d\eta_i)/V(\mu_i)$ is 1. $\square$

Theorem 8.30 (Iteratively reweighted least squares). Let

$$W_i = \frac{w_i}{\phi V(\mu_i)} \left(\frac{d\mu_i}{d\eta_i} \right)^2, \quad z_i = \eta_i + \frac{y_i - \mu_i}{d\mu_i/d\eta_i}, \quad (23)$$

both evaluated at the current $\boldsymbol{\beta}^{(m)}$. The Fisher scoring step is

$$\boldsymbol{\beta}^{(m+1)} = \boldsymbol{\beta}^{(m)} + \mathcal{I}(\boldsymbol{\beta}^{(m)})^{-1} \nabla \ell(\boldsymbol{\beta}^{(m)}).$$

It equals the weighted least squares fit $(\mathbf{X}'\mathbf{W}\mathbf{X})^{-1}\mathbf{X}'\mathbf{W}\mathbf{z}$, with $\mathbf{W} = \text{diag}(W_i)$. The step does not involve ϕ , which cancels.

Proof. By Theorem 8.29, $\mathcal{I} = \mathbf{X}'\mathbf{W}\mathbf{X}$ and $\nabla\ell = \sum_i W_i(z_i - \eta_i)\mathbf{x}_i = \mathbf{X}'\mathbf{W}(\mathbf{z} - \boldsymbol{\eta})$, since $W_i(z_i - \eta_i) = W_i(y_i - \mu_i)/(\text{d}\mu_i/\text{d}\eta_i)$ and substituting W_i from (23) gives the score summand. Hence

$$\boldsymbol{\beta}^{(m)} + (\mathbf{X}'\mathbf{W}\mathbf{X})^{-1}\mathbf{X}'\mathbf{W}(\mathbf{z} - \mathbf{X}\boldsymbol{\beta}^{(m)}) = (\mathbf{X}'\mathbf{W}\mathbf{X})^{-1}\mathbf{X}'\mathbf{W}\mathbf{z},$$

using $\boldsymbol{\eta} = \mathbf{X}\boldsymbol{\beta}^{(m)}$. Both $\mathbf{W}$ and $\mathcal{I}$ carry the factor $1/\phi$, which cancels between them. $\square$

3.3 Deviance and residuals

Definition 8.31 (Deviance). Let $\tilde{\theta}_i$ solve $b'(\tilde{\theta}_i) = y_i$, the **saturated** fit, and let $\hat{\theta}_i$ be the fitted natural parameters. The **deviance** is

$$D = 2 \sum_{i=1}^n w_i \{y_i(\tilde{\theta}_i - \hat{\theta}_i) - b(\tilde{\theta}_i) + b(\hat{\theta}_i)\} = \sum_{i=1}^n d_i, \quad (24)$$

and the **scaled deviance** is $D/\phi = 2\{\ell_{\text{sat}} - \ell(\hat{\boldsymbol{\beta}})\}$.

Proposition 8.32 (Non-negativity and special cases). Every $d_i \geq 0$, with equality only when $\hat{\mu}_i = y_i$. For the three standard families,

$$\text{normal: } D = \sum_i w_i (y_i - \hat{\mu}_i)^2, \quad (25)$$

$$\text{Poisson: } D = 2 \sum_i \left\{ y_i \ln \frac{y_i}{\hat{\mu}_i} - (y_i - \hat{\mu}_i) \right\}, \quad (26)$$

$$\text{gamma: } D = 2 \sum_i \left\{ -\ln \frac{y_i}{\hat{\mu}_i} + \frac{y_i - \hat{\mu}_i}{\hat{\mu}_i} \right\}, \quad (27)$$

with the convention $0 \ln 0 = 0$.

Proof. Put $h(\theta) = y_i \theta - b(\theta)$, so $d_i = 2w_i \{h(\tilde{\theta}_i) - h(\hat{\theta}_i)\}$. Now $h'(\theta) = y_i - b'(\theta)$ vanishes at $\tilde{\theta}_i$ and $h'' = -b'' < 0$, so h is strictly concave with its maximum at $\tilde{\theta}_i$; hence $d_i \geq 0$ with equality only at $\hat{\theta}_i = \tilde{\theta}_i$, that is $\hat{\mu}_i = y_i$.

Normal: $b(\theta) = \theta^2/2$, $\theta = \mu$, so $d_i = 2w_i\{y_i(y_i - \hat{\mu}_i) - \frac{1}{2}y_i^2 + \frac{1}{2}\hat{\mu}_i^2\} = w_i(y_i - \hat{\mu}_i)^2$. Poisson: $b(\theta) = e^\theta$, $\theta = \ln \mu$, $w_i = 1$, so $d_i = 2\{y_i \ln y_i - y_i \ln \hat{\mu}_i - y_i + \hat{\mu}_i\}$, which is (26). Gamma: $b(\theta) = -\ln(-\theta)$ and $\theta = -1/\mu$, so $d_i = 2w_i\{-y_i/y_i + y_i/\hat{\mu}_i + \ln y_i - \ln \hat{\mu}_i\}$ with $w_i = 1$, which is (27). $\square$

Definition 8.33 (Residuals). The **Pearson** and **deviance** residuals are

$$r_i^P = \frac{\sqrt{w_i}(y_i - \hat{\mu}_i)}{\sqrt{V(\hat{\mu}_i)}}, \quad r_i^D = \text{sign}(y_i - \hat{\mu}_i)\sqrt{d_i}, \quad (28)$$

so that $\sum_i (r_i^P)^2 = X^2$, the Pearson statistic, and $\sum_i (r_i^D)^2 = D$.

Remark. Raw residuals $y_i - \hat{\mu}_i$ are useless for diagnostics in a GLM because their variance $\phi V(\mu_i)/w_i$ moves with the fit: a plot of them against $\hat{\mu}_i$ shows the variance function, not a lack of fit. Both residuals in (28) are constructed to have variance approximately ϕ ; the deviance residual is usually the more nearly symmetric of the two for skewed families, and it is the one whose squares add to a quantity with a familiar interpretation.

3.4 The Tweedie family

Definition 8.34 (Tweedie). The **Tweedie** family with index p is the linear exponential family with variance function $V(\mu) = \mu^p$. For $p \notin \{1, 2\}$,

$$\theta = \frac{\mu^{1-p}}{1-p}, \quad b(\theta) = \frac{\mu^{2-p}}{2-p}, \quad (29)$$

μ being written for $\{(1-p)\theta\}^{1/(1-p)}$. The cases $p = 0, 1, 2, 3$ are the normal, Poisson, gamma and inverse Gaussian.

Proposition 8.35. The pair (29) satisfies $b'(\theta) = \mu$ and $b''(\theta) = \mu^p$.

Proof. From $\theta = \mu^{1-p}/(1-p)$, $d\theta/d\mu = \mu^{-p}$. Then $db/d\mu = \mu^{1-p}$, so $b'(\theta) = (db/d\mu)(d\mu/d\theta) = \mu^{1-p}\mu^p = \mu$, and differentiating once more, $b''(\theta) = (d\mu/d\theta) = \mu^p$. $\square$

Lemma 8.36 (Sum of gammas). If $X \sim \text{Gamma}(\alpha, \gamma)$ and $Y \sim \text{Gamma}(\beta, \gamma)$ are independent, with density $\gamma^\alpha u^{\alpha-1} e^{-\gamma u} / \Gamma(\alpha)$ on $(0, \infty)$, then $X + Y \sim \text{Gamma}(\alpha + \beta, \gamma)$.

Proof. First, $\Gamma(\alpha)\Gamma(\beta) = \Gamma(\alpha + \beta)B(\alpha, \beta)$ with $B(\alpha, \beta) = \int_0^1 t^{\alpha-1}(1-t)^{\beta-1}dt$: substituting $u = st$, $v = s(1-t)$ with Jacobian s in the double integral and using Fubini–Tonelli,

$$\Gamma(\alpha)\Gamma(\beta) = \int_0^\infty \int_0^\infty u^{\alpha-1}v^{\beta-1}e^{-u-v}du dv = \int_0^\infty s^{\alpha+\beta-1}e^{-s}ds \int_0^1 t^{\alpha-1}(1-t)^{\beta-1}dt.$$

Now convolve: for $s > 0$,

$$f_{X+Y}(s) = \frac{\gamma^{\alpha+\beta}e^{-\gamma s}}{\Gamma(\alpha)\Gamma(\beta)} \int_0^s u^{\alpha-1}(s-u)^{\beta-1}du = \frac{\gamma^{\alpha+\beta}e^{-\gamma s}s^{\alpha+\beta-1}}{\Gamma(\alpha)\Gamma(\beta)}B(\alpha, \beta),$$

substituting $u = st$, and the first display turns the constant into $\gamma^{\alpha+\beta}/\Gamma(\alpha + \beta)$. □

Theorem 8.37 (Compound Poisson–gamma). Let $N \sim \text{Pois}(\lambda)$ and let $Y_1, Y_2, \dots$ be independent $\text{Gamma}(\alpha, \gamma)$ variables independent of N , and set $S = \sum_{j \leq N} Y_j$, an empty sum being 0.

Given $\mu > 0$, $\phi > 0$ and $1 < p < 2$, put

$$\lambda = \frac{\mu^{2-p}}{\phi(2-p)}, \quad \alpha = \frac{2-p}{p-1}, \quad \frac{1}{\gamma} = \phi(p-1)\mu^{p-1}. \quad (30)$$

Then all three are positive,

$$\mathbf{E}[S] = \mu, \quad \text{var}(S) = \phi\mu^p, \quad \mathbf{P}\{S = 0\} = e^{-\lambda}, \quad (31)$$

and for $y > 0$ the law of S has density

$$f_S(y) = \sum_{k=1}^{\infty} e^{-\lambda} \frac{\lambda^k}{k!} \frac{\gamma^{k\alpha}}{\Gamma(k\alpha)} y^{k\alpha-1} e^{-\gamma y} \quad (32)$$

with respect to Lebesgue measure.

Proof. Positivity is clear from $1 < p < 2$. By the random-sum corollary of the prerequisites,

with $\mathbf{E}[Y_1] = \alpha/\gamma$ and $\text{var}(Y_1) = \alpha/\gamma^2$,

$$\mathbf{E}[S] = \mathbf{E}[N] \mathbf{E}[Y_1] = \lambda \frac{\alpha}{\gamma}, \quad \text{var}(S) = \mathbf{E}[N] \text{var}(Y_1) + \text{var}(N)(\mathbf{E} Y_1)^2 = \frac{\lambda \alpha}{\gamma^2}(1 + \alpha),$$

using $\mathbf{E}[N] = \text{var}(N) = \lambda$. Now $\alpha/\gamma = \{(2-p)/(p-1)\}\phi(p-1)\mu^{p-1} = \phi(2-p)\mu^{p-1}$, so $\mathbf{E}[S] = \{\mu^{2-p}/(\phi(2-p))\}\phi(2-p)\mu^{p-1} = \mu$. Also $1 + \alpha = 1/(p-1)$, so

$$\text{var}(S) = \left(\frac{\lambda \alpha}{\gamma}\right) \frac{1 + \alpha}{\gamma} = \mu \cdot \frac{1}{p-1} \cdot \phi(p-1)\mu^{p-1} = \phi\mu^p.$$

Since every $Y_j > 0$ almost surely, $S = 0$ exactly when $N = 0$, of probability $e^{-\lambda}$.

For the density, condition on $N = k \geq 1$: by Lemma 8.36 and induction, $Y_1 + \cdots + Y_k \sim \text{Gamma}(k\alpha, \gamma)$. Hence for a Borel $B \subset (0, \infty)$,

$$\mathbf{P}\{S \in B\} = \sum_{k \geq 1} \mathbf{P}\{N = k\} \int_B \frac{\gamma^{k\alpha}}{\Gamma(k\alpha)} y^{k\alpha-1} e^{-\gamma y} dy,$$

and interchanging the sum and the integral, legitimate by Tonelli since every term is non-negative, exhibits (32) as the density. $\square$

Remark. The Tweedie with $1 < p < 2$ is the natural model for an insurance loss on a policy: a point mass at zero for the policies with no claim, and a continuous positive part for the aggregate of those that do. The frequency and severity are not separately identified from the aggregate alone; what the fit delivers is the mean μ and the pair (ϕ, p) , and p interpolates between the Poisson at $p = 1$ and the gamma at $p = 2$. Fitting proceeds by maximising the likelihood over β for a grid of p , since p enters $V(\mu) = \mu^p$ and not the linear predictor.

4 Computation: Formulas Against Package Output

FREES §12.2, §13.4.

Four data sets are used. Counts: **Bikeshare** from **ISLR2**, hourly bicycle rentals. Counts with exposure: **Insurance** from **MASS**, third-party motor claims with the number of policy holders as the exposure. Excess zeros: **bioChemists** from **pscl**. Severities: the **AutoCollision** table of FREES §13.2, entered from the printed page.

```
1 suppressMessages({library(ISLR2); library(MASS); library(pscl)
2                     library(statmod); library(tweedie)})
3 d <- Bikeshare; n <- nrow(d); y <- d$bikers
```

```

4 ctl <- glm.control(epsilon = 1e-14, maxit = 200)
5 c(n = n, mean = mean(y), variance = var(y), zeros = sum(y == 0))

```

```

##          n      mean variance    zeros
## 8645.0    143.8  17901.9      0.0

```

4.1 Score, information, IRLS (§1.3)

```

1 g <- glm(bikers ~ mnth + hr + workingday + temp + weathersit,
2         data = d, family = poisson, control = ctl)
3 X <- model.matrix(g); b <- coef(g); mu <- fitted(g)
4 U <- drop(crossprod(X, y - mu))           # score:  $X'(y - \mu)$ 
5 Info <- t(X) %*% (mu * X)                 # information:  $X'WX$ 
6 c(parameters = length(b), max_abs_score = max(abs(U)),
7   se_manual_vs_glm = max(abs(sqrt(diag(solve(Info)))) -
8                           summary(g)$coefficients[, 2])),
9   smallest_se = min(summary(g)$coefficients[, 2]))

```

```

##      parameters    max_abs_score se_manual_vs_glm
##      4.000e+01      1.214e-09      4.651e-14
##      smallest_se
##      1.955e-03

```

```

1 c(total_observed = sum(y), total_fitted = sum(mu),
2   workingday_observed = sum(y[d$workingday == 1]),
3   workingday_fitted = sum(mu[d$workingday == 1]))

```

```

##          total_observed          total_fitted workingday_observed
##                1243103                1243103                856264
## workingday_fitted
##                856264

```

```

1 irls <- function(X, y, off = rep(0, length(y)), tol = 1e-12) {
2    b <- c(log(mean(y)), rep(0, ncol(X) - 1))
3    repeat {
4      eta <- drop(X %*% b) + off; m <- exp(eta); w <- m
5      z <- eta - off + (y - m)/w
6      bn <- drop(solve(t(X) %*% (w*X), crossprod(X, w*z)))
7      if (max(abs(bn - b)) < tol) return(bn)
8      b <- bn
9    }
10 }
11 c(irls_vs_glm = max(abs(irls(X, y) - b)), coefficient_scale = max(abs(b)))

```



```
##          irls_vs_glm coefficient_scale
##          1.341e-13          2.694e+00
```

4.2 Exposure and the offset (§1.2)

```
1 ins <- MASS::Insurance
2 gi <- glm(Claims ~ District + Group + Age + offset(log(Holders)),
3         data = ins, family = poisson, control = ctl)
4 c(rows = nrow(ins), observed_claims = sum(ins$Claims),
5   fitted_claims = sum(fitted(gi)), deviance = deviance(gi),
6   residual_df = df.residual(gi),
7   pearson_over_df = sum(residuals(gi, "pearson")^2)/df.residual(gi))
```

```
##          rows observed_claims  fitted_claims
##          64.0000      3151.0000      3151.0000
##          deviance      residual_df pearson_over_df
##          51.4200      54.0000      0.9005
```

```
1 round(exp(coef(gi)), 4)
```

```
## (Intercept)  District2  District3  District4  Group.L
##          0.1636      1.0262      1.0393      1.2639      1.5368
```

##	Group.Q	Group.C	Age.L	Age.Q	Age.C
##	1.0046	0.9711	0.6741	0.9996	0.9834

The Pearson statistic per degree of freedom is close to 1, so the Poisson variance assumption survives on this data.

4.3 Overdispersion (§2.1–§2.2)

```

1 phi <- sum((y - mu)^2/mu)/(n - length(b))
2 gq <- glm(bikers ~ mnth + hr + workingday + temp + weathersit,
3           data = d, family = quasipoisson, control = ctl)
4 c(phi_pearson = phi, phi_reported = summary(gq)$dispersion,
5   deviance_over_df = deviance(g)/(n - length(b)),
6   coefficients_identical = max(abs(coef(gq) - b)),
7   se_ratio_minus_sqrt_phi =
8     max(abs(summary(gq)$coefficients[, 2]/summary(g)$coefficients[, 2] -
9         sqrt(phi))),
10  sqrt_phi = sqrt(phi))

```

##	phi_pearson	phi_reported
##	2.560e+01	2.560e+01
##	deviance_over_df	coefficients_identical

##	2.650e+01	0.000e+00
## se_ratio_minus_sqrt_phi		sqrt_phi
##	9.059e-13	5.060e+00

Theorem 8.14 is exact arithmetic here: the coefficients agree to the last bit and every standard error is multiplied by the same $\sqrt{\hat{\phi}} = 5.06$. The two dispersion estimates of §2.1 differ, 25.60 against 26.50.

```

1 Ti <- (y - mu)^2 - y
2 c(alpha_v_equals_mu = unname(coef(lm(Ti ~ mu - 1))),
3   alpha_v_equals_mu2 = unname(coef(lm(Ti ~ I(mu^2) - 1))),
4   implied_theta = 1/unname(coef(lm(Ti ~ I(mu^2) - 1))))

```

##	alpha_v_equals_mu	alpha_v_equals_mu2	implied_theta
##	33.87917	0.09821	10.18211

Both slopes are far from zero, which is the point of the test. Neither reproduces the corresponding maximum likelihood quantity: $1 + \hat{\alpha} = 34.9$ against $\hat{\phi} = 25.6$, and $1/\hat{\alpha} = 10.2$ against $\hat{\theta} = 3.76$. The regressions are unweighted, so observations with large $\hat{\mu}_i$, whose T_i have far the largest variance, dominate them; the test is a diagnostic for the null, not an estimator to report.

```

1 G <- t(X) %*% (mu * X); Fm <- t(X) %*% ((y - mu)^2 * X)

```

```

2 sw <- sqrt(diag(solve(G) %*% Fm %*% solve(G)))
3 r <- sw/summary(g)$coefficients[, 2]
4 c(smallest_ratio = min(r), largest_ratio = max(r),
5   quasi_poisson_ratio = sqrt(phi))

```

```

##          smallest_ratio          largest_ratio quasi_poisson_ratio
##                0.1564                7.0656                5.0595

```

Quasi-Poisson forces one ratio for every coefficient; the sandwich finds ratios from 0.16 to 7.07. The variance is not a constant multiple of the mean on this data, and the quasi-Poisson standard error is too large for some coefficients and too small for others.

4.4 The negative binomial (§2.3)

```

1 nb <- glm.nb(bikers ~ mnth + hr + workingday + temp + weathersit,
2             data = d, control = ctl)
3 mun <- fitted(nb)
4 c(theta = nb$theta, se_theta = nb$SE.theta,
5   poisson_AIC = AIC(g), negbin_AIC = AIC(nb),
6   poisson_pearson_over_df = phi,
7   negbin_pearson_over_df =

```

```
8 sum((y - mun)^2/(mun + mun^2/nb$theta))/(n - length(b) - 1))
```

```
##                theta                se_theta
##                3.75987                0.06165
##      poisson_AIC      negbin_AIC
##      281158.77871      89491.40732
## poisson_pearson_over_df  negbin_pearson_over_df
##                25.59892                0.89997
```

```
1 set.seed(5); N <- 400000; th <- 2.5; mu0 <- 3
2 lam <- rgamma(N, shape = th, rate = th/mu0)      # mean mu0, var mu0^2/th
3 ym <- rpois(N, lam)
4 round(rbind(simulated = as.numeric(table(factor(ym, levels = 0:8))/N),
5           negative_binomial = dnbinom(0:8, size = th, mu = mu0)), 5)
```

```
##                [,1] [,2] [,3] [,4] [,5] [,6]
## simulated      0.1402 0.189 0.1812 0.1496 0.1103 0.07897
## negative_binomial 0.1393 0.190 0.1813 0.1484 0.1113 0.07890
##                [,7] [,8] [,9]
## simulated      0.05401 0.03533 0.02314
## negative_binomial 0.05379 0.03563 0.02308
```

```
1 c(simulated_mean = mean(ym), theory = mu0,  
2   simulated_var = var(ym), theory_var = mu0 + mu0^2/th)
```

##	simulated_mean	theory	simulated_var	theory_var
##	2.999	3.000	6.619	6.600

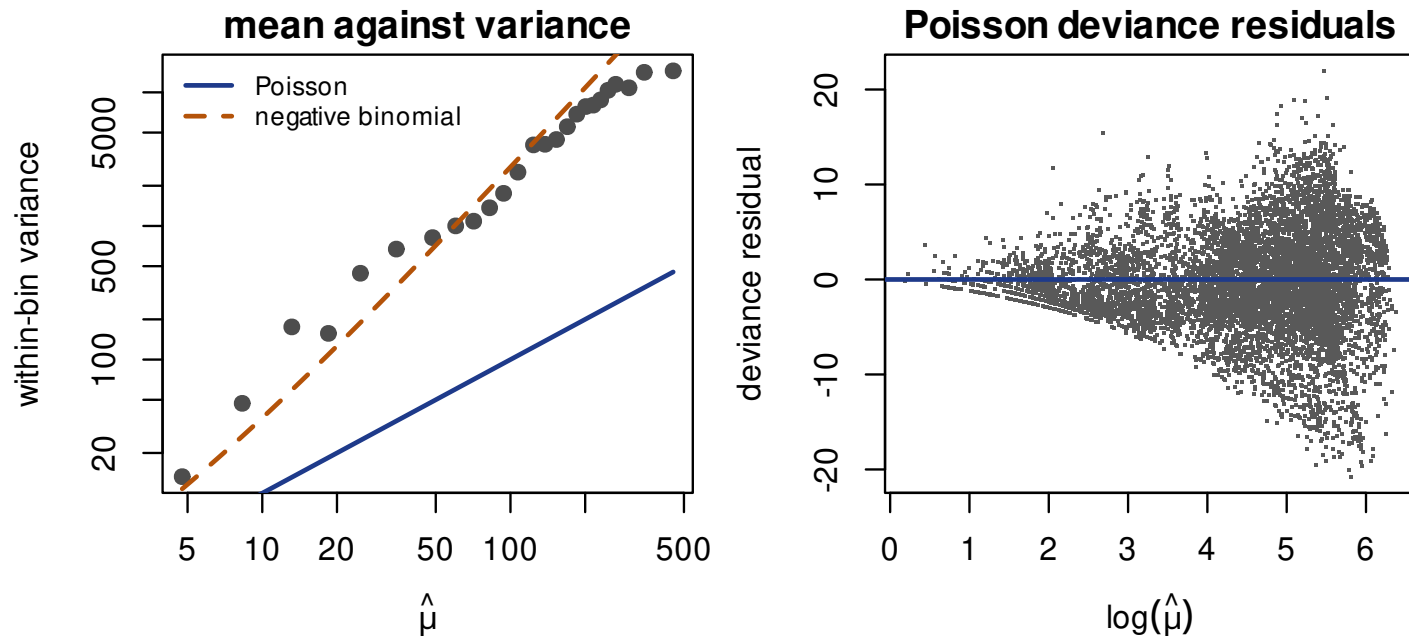


Figure 8.1: Left: the within-bin variance of **bikers** against the fitted Poisson mean, in 25 bins of equal size, with the Poisson line $V = \mu$ and the fitted negative binomial curve $V = \mu + \mu^2/\hat{\theta}$. Right: deviance residuals of the Poisson fit, which should have variance about 1 and instead reach ± 20 .

4.5 Excess zeros (§2.4)

```

1 data("bioChemists", package = "pscl"); bc <- bioChemists
2 f0 <- art ~ fem + mar + kid5 + phd + ment
3 gp <- glm(f0, data = bc, family = poisson, control = ctl)
4 nb2 <- glm.nb(f0, data = bc, control = ctl)
5 zi <- zeroinfl(art ~ fem + mar + kid5 + phd + ment | 1, data = bc)
6 hu <- hurdle(art ~ fem + mar + kid5 + phd + ment | 1, data = bc)
7 c(n = nrow(bc), mean = mean(bc$art), variance = var(bc$art))

```

```

##           n      mean variance
## 915.000    1.693    3.710

```

```

1 c(poisson = AIC(gp), negbin = AIC(nb2), zip = AIC(zi), hurdle = AIC(hu))

```

```

## poisson  negbin      zip  hurdle
##    3314    3136    3256    3293

```

```

1 c(observed_zeros = sum(bc$art == 0),
2   poisson_expected = sum(dpois(0, fitted(gp))),
3   negbin_expected = sum(dnbinom(0, size = nb2$theta, mu = fitted(nb2))),
4   zip_expected = sum(predict(zi, type = "prob")[, 1]),

```



```
5 hurdle_expected = sum(predict(hu, type = "prob")[, 1]))
```

```
## observed_zeros poisson_expected negbin_expected
##          275.0          191.4          277.8
## zip_expected hurdle_expected
##          264.4          275.0
```

The Poisson expects 191 zeros where 275 occur. The hurdle reproduces 275 exactly, which is Corollary 8.23, and the negative binomial comes close without being constructed to.

```
1 Xz <- model.matrix(~ fem + mar + kid5 + phd + ment, data = bc)
2 cf <- coef(zi)
3 mz <- drop(exp(Xz %*% cf[grep("^count_", names(cf))]))
4 p0 <- plogis(cf[[grep("^zero_", names(cf))]])
5 a <- bc$art
6 ll <- sum(ifelse(a == 0, log(p0 + (1 - p0)*dpois(0, mz)),
7               log(1 - p0) + dpois(a, mz, log = TRUE)))
8 c(manual_loglik = ll, zeroinfl_loglik = as.numeric(logLik(zi)), pi_hat = p0,
9   zip_mean = mean((1 - p0)*mz), sample_mean = mean(a),
10  zip_variance = mean((1 - p0)*mz + p0*(1 - p0)*mz^2),
11  sample_variance = var(a))
```

```
##      manual_loglik zeroinfl_loglik          pi_hat
##      -1620.7840      -1620.7840          0.1569
##      zip_mean      sample_mean      zip_variance
##      1.6711          1.6929          2.2446
## sample_variance
##      3.7097
```

Theorem 8.21 gives a variance of 2.25 against a sample variance of 3.71: zero inflation produces overdispersion but not enough of it here, so the negative binomial has the smaller AIC.

```
1 cfh <- coef(hu)
2 mh  <- drop(exp(Xz %*% cfh[grepl("^count_", names(cfh))]))
3 p1  <- plogis(cfh[[grepl("^zero_", names(cfh))]])
4 llh <- sum(ifelse(a == 0, log(1 - p1),
5                   log(p1) + dpois(a, mh, log = TRUE) -
6                   log(1 - dpois(0, mh))))
7 c(manual_loglik = llh, hurdle_loglik = as.numeric(logLik(hu)),
8   fitted_p_positive = p1, observed_fraction_positive = mean(a > 0))
```

```
##      manual_loglik      hurdle_loglik
##      -1639.3974      -1639.3974
##      fitted_p_positive observed_fraction_positive
```

```
##                                0.6995                                0.6995
```

4.6 The exponential family (§3.1, §3.3)

```
1 fam <- list(
2   normal    = list(b = function(t) t^2/2,          th = 1.7),
3   Bernoulli = list(b = function(t) log(1 + exp(t)), th = qlogis(0.3)),
4   Poisson   = list(b = function(t) exp(t),          th = log(2.5)),
5   gamma     = list(b = function(t) -log(-t),        th = -1/4),
6   invGauss  = list(b = function(t) -sqrt(-2*t),     th = -1/(2*3^2)))
7 h <- 1e-5
8 round(t(sapply(fam, function(f) {
9   t0 <- f$th
10   c(theta = t0, b_prime = (f$b(t0+h) - f$b(t0-h))/(2*h),
11     b_second = (f$b(t0+h) - 2*f$b(t0) + f$b(t0-h))/h^2 })), 5)
```

```
##          theta b_prime b_second
## normal      1.70000    1.7     1.00
## Bernoulli -0.84730    0.3     0.21
## Poisson     0.91629    2.5     2.50
## gamma      -0.25000    4.0    16.00
## invGauss   -0.05556    3.0    27.00
```

The targets are $\mu = 1.7$ with $V = 1$; $\mu = 0.3$ with $V = \mu(1 - \mu) = 0.21$; $\mu = 2.5$ with $V = \mu$; $\mu = 4$ with $V = \mu^2 = 16$; and $\mu = 3$ with $V = \mu^3 = 27$.

```

1 di <- 2*(ifelse(y == 0, 0, y*log(y/mu)) - (y - mu))    # Poisson deviance terms
2 dr <- sign(y - mu)*sqrt(pmax(di, 0))
3 c(deviance_manual = sum(di), deviance_glm = deviance(g),
4   residuals_max_diff = max(abs(dr - residuals(g, "deviance"))),
5   pearson_manual = sum((y - mu)^2/mu),
6   pearson_glm = sum(residuals(g, "pearson")^2))

```

```

##      deviance_manual      deviance_glm residuals_max_diff
##              228041              228041              0
##      pearson_manual      pearson_glm
##              220279              220279

```

4.7 A gamma severity model (§3.2)

The `AutoCollision` table of FREES §13.2: average claim severity in pounds by driver age group and vehicle use, with the claim count as the weight w_i .

```

1 sev <- c(250.48,274.78,244.52,797.80, 213.71,298.60,298.13,362.23,
2         250.57,248.56,297.90,342.31, 229.09,228.48,293.87,367.46,

```

```

3      153.62,201.67,238.21,256.21, 208.59,202.80,236.06,352.49,
4      207.57,202.67,253.63,340.56, 192.00,196.33,259.79,342.58)
5 cnt <- c(21,40,23,5, 63,171,92,44, 140,343,318,129, 123,448,361,169,
6      151,479,381,166, 245,970,719,304, 266,859,504,162, 260,578,312,96)
7 ac <- data.frame(sev, cnt,
8   age = factor(rep(c("17-20","21-24","25-29","30-34","35-39","40-49",
9   "50-59","60+"), each = 4)),
10  use = factor(rep(c("Pleasure","DriveShort","DriveLong","Business"), 8),
11    levels = c("Pleasure","DriveShort","DriveLong","Business")))
12 gg <- glm(sev ~ age + use, data = ac, family = Gamma(link = "log"),
13   weights = cnt)
14 c(cells = nrow(ac), total_claims = sum(cnt),
15   dispersion = summary(gg)$dispersion, deviance = deviance(gg),
16   residual_df = df.residual(gg))

```

```

##      cells total_claims  dispersion  deviance
##      32.000    8942.000      1.543      31.838
## residual_df
##      21.000

```

```
1 round(exp(coef(gg)), 4)
```

##	(Intercept)	age21-24	age25-29	age30-34
##	254.8971	0.9953	0.9227	0.8842
##	age35-39	age40-49	age50-59	age60+
##	0.7119	0.7702	0.7820	0.7650
##	useDriveShort	useDriveLong	useBusiness	
##	1.0418	1.2639	1.6441	

The exponentiated coefficients are the multiplicative rating factors of Proposition 8.5(iv): a base of \$254.90 for a 17–20 driver using the vehicle for pleasure, times 1.644 for business use and times 0.765 for the oldest age group. FREES reports 254.90 and 419.07 for the first row of its fitted table, against 254.90 and $254.90 \times 1.6441 = 419.08$ here.

4.8 Tweedie (§3.4)

```

1 set.seed(11); N <- 300000; mu0 <- 4; phi0 <- 2.5; p0 <- 1.6
2 lam <- mu0^(2 - p0)/(phi0*(2 - p0))
3 al <- (2 - p0)/(p0 - 1)
4 gam <- 1/(phi0*(p0 - 1)*mu0^(p0 - 1))
5 Nn <- rpois(N, lam)
6 S <- vapply(Nn, function(k) if (k == 0) 0 else
7           sum(rgamma(k, shape = al, rate = gam)), numeric(1))

```

```

8 c(lambda = lam, alpha = al, rate = gam,
9   simulated_mean = mean(S), theory_mean = mu0,
10  simulated_var = var(S), theory_var = phi0*mu0^p0,
11  simulated_P0 = mean(S == 0), theory_P0 = exp(-lam))

```

```

##          lambda          alpha          rate simulated_mean
##          1.7411          0.6667          0.2902          4.0104
## theory_mean simulated_var      theory_var      simulated_P0
##          4.0000          23.0605          22.9740          0.1751
##          theory_P0
##          0.1753

```

```

1 yv <- c(1, 4, 10)
2 series <- sapply(yv, function(u)
3   sum(sapply(1:200, function(k)
4     exp(-lam)*lam^k/factorial(k)*dgamma(u, shape = k*al, rate = gam))))
5 rbind(y = yv, series = series,
6       dtweedie = dtweedie(yv, xi = p0, mu = mu0, phi = phi0))

```

```

##          [,1]      [,2]      [,3]
## y          1.0000  4.00000 10.00000

```

```
## series    0.1278 0.07067  0.02277
## dtweedie 0.1278 0.07067  0.02277
```

```
1 set.seed(12); nt <- 4000
2 xt <- runif(nt, -1, 1)
3 yt <- rtweedie(nt, xi = p0, mu = exp(1 + 0.8*xt), phi = phi0)
4 gt <- glm(yt ~ xt, family = tweedie(var.power = p0, link.power = 0))
5 c(true_intercept = 1, true_slope = 0.8, fitted = coef(gt),
6   zero_fraction = mean(yt == 0),
7   dispersion = summary(gt)$dispersion, true_phi = phi0)
```

```
##      true_intercept      true_slope fitted.(Intercept)
##           1.0000           0.8000           1.0265
##      fitted.xt      zero_fraction      dispersion
##           0.8064           0.2320           2.4060
##      true_phi
##           2.5000
```

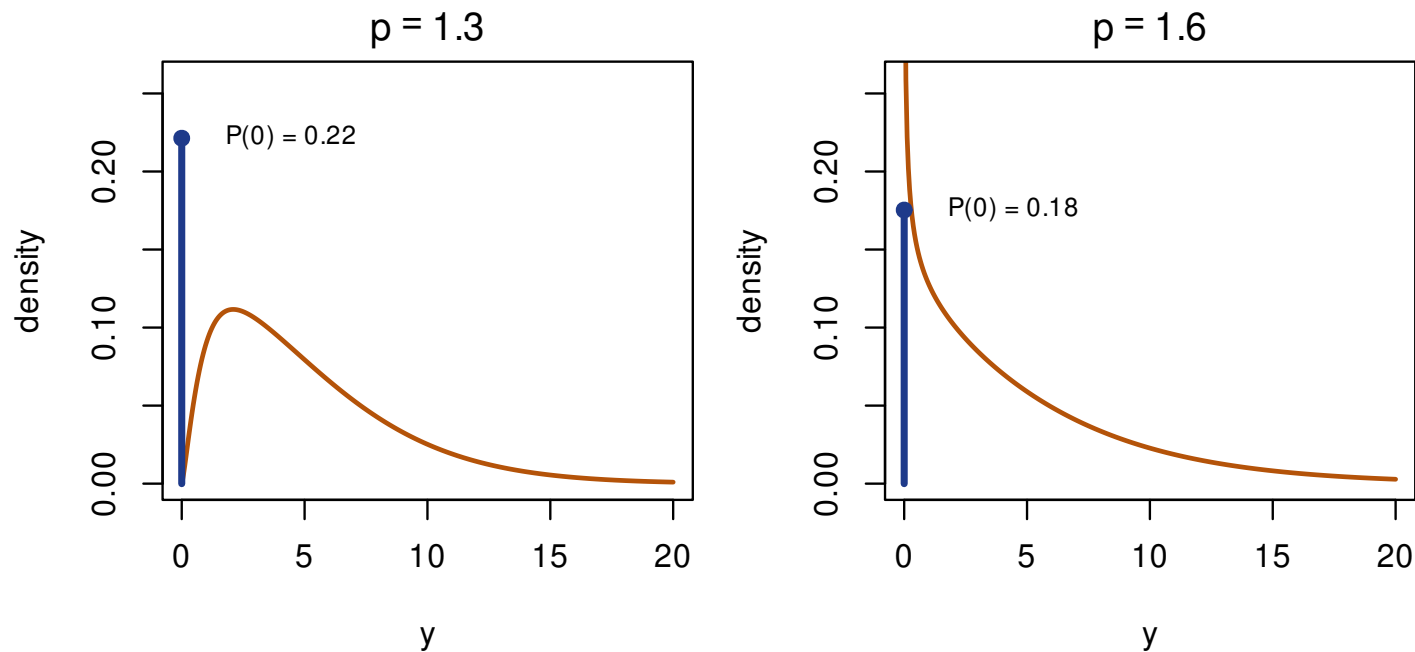



Figure 8.2: Tweedie densities at $\mu = 4$, $\phi = 2.5$ and two indices. The spike at the origin is the point mass $e^{-\lambda}$ of Theorem 8.37; the curve is the density (32) of the positive part. Larger p puts more mass at zero and makes the positive part more skewed.

5 Pitfalls

5.1 Poisson standard errors on overdispersed counts

```
1 j <- which(names(b) == "temp")
2 c(coefficient = unname(b[j]),
3   poisson_se = summary(g)$coefficients[j, 2],
4   quasi_se = summary(gq)$coefficients[j, 2],
5   sandwich_se = unname(sw[j]),
6   poisson_z = summary(g)$coefficients[j, 3],
7   quasi_t = summary(gq)$coefficients[j, 3])
```

```
## coefficient   poisson_se    quasi_se sandwich_se   poisson_z
##      0.78529      0.01148      0.05806      0.06499      68.43424
##      quasi_t
##      13.52579
```

Diagnostic. A Pearson statistic per degree of freedom far from 1, together with z statistics in the hundreds. Every coefficient looks decisive because the model is asserting that a count with mean 200 has standard deviation 14. **Fix.** Estimate the dispersion and use Theorem 8.14, or fit a model whose variance function can match the data, or use the sandwich of Proposition 8.15. Note also that

a likelihood ratio statistic must be divided by $\hat{\phi}$ before it is referred to χ^2 : it is a difference of criterion values, not of log-likelihoods.

5.2 Exposure entered as a covariate

An offset asserts that the mean is proportional to the exposure. Estimating a coefficient for $\ln E_i$ instead relaxes that to $\mu_i \propto E_i^\gamma$, which is a different model and is rarely what is meant.

```

1 g2 <- glm(Claims ~ District + Group + Age + log(Holders), data = ins,
2           family = poisson, control = ctl)
3 c(gamma_hat = unname(coef(g2)["log(Holders)"]),
4   se = summary(g2)$coefficients["log(Holders)", 2],
5   LR_against_offset = deviance(gi) - deviance(g2),
6   p_value = pchisq(deviance(gi) - deviance(g2), 1, lower.tail = FALSE),
7   effect_of_doubling_exposure_offset = 2,
8   effect_of_doubling_exposure_covariate =
9     2~unname(coef(g2)["log(Holders)"]))

```

```

##                gamma_hat
##                1.2017
##                se
##                0.1441

```

```
##                LR_against_offset
##                1.9700
##                p_value
##                0.1604
## effect_of_doubling_exposure_offset
##                2.0000
## effect_of_doubling_exposure_covariate
##                2.3001
```

Diagnostic. A coefficient on $\ln E_i$ reported among the others, and rates that are not proportional to exposure. Here $\hat{\gamma} = 1.20$ and the restriction $\gamma = 1$ is not rejected, yet doubling the exposure multiplies the fitted claims by 2.30 rather than by 2. **Fix.** Use `offset(log(E))` when the model is a rate. If proportionality is genuinely in doubt, fit both and test, and say which was used.

5.3 Testing Poisson against negative binomial by a likelihood ratio

The Poisson is the $\theta \rightarrow \infty$ limit of the negative binomial (Corollary 8.18), not an interior point of its parameter space, so the likelihood ratio theorem of Unit 7 does not apply: its proof requires the null value to be interior, which is where the Taylor expansion of the score is taken.

```

1 LR <- 2*(as.numeric(logLik(nb2)) - as.numeric(logLik(gp)))
2 c(LR = LR, naive_p_chisq_1 = pchisq(LR, 1, lower.tail = FALSE),
3   poisson_AIC = AIC(gp), negbin_AIC = AIC(nb2),
4   theta_hat = nb2$theta, se_theta = nb2$SE.theta)

```

##	LR	naive_p_chisq_1	poisson_AIC
##	1.802e+02	4.392e-41	3.314e+03
##	negbin_AIC	theta_hat	se_theta
##	3.136e+03	2.264e+00	2.712e-01

Diagnostic. A p -value quoted from χ_1^2 for a hypothesis on the boundary of the parameter space.

Fix. Compare by an information criterion, which needs no null distribution, or use the moment test of Proposition 8.11, whose null hypothesis $\alpha = 0$ is interior to $\alpha \in \mathbb{R}$ once under- as well as overdispersion is allowed.

5.4 Reading the deviance as a goodness-of-fit statistic

The rule that the deviance of a correct model should be about $n - p$ is an asymptotic statement in which each mean grows, not in which n grows. Below, the model is exactly correct and only the size of the means changes.

```

1 set.seed(21); nn <- 2000
2 t(sapply(c(0.3, 1, 5, 50), function(m) {
3   D <- replicate(200, deviance(glm(rpois(nn, m) ~ 1, family = poisson)))
4   c(mu = m, mean_deviance = mean(D), n_minus_p = nn - 1,
5     sd_deviance = sd(D), chi_square_sd = sqrt(2*(nn - 1))) })))

```

##		mu	mean_deviance	n_minus_p	sd_deviance	chi_square_sd
##	[1,]	0.3	1683	1999	38.46	63.23
##	[2,]	1.0	2290	1999	53.56	63.23
##	[3,]	5.0	2092	1999	69.89	63.23
##	[4,]	50.0	2003	1999	68.27	63.23

Diagnostic. A deviance far from its degrees of freedom on data whose counts are mostly 0 and 1, read as evidence of misspecification. **Fix.** With small means, compare deviances between nested models, which is a valid likelihood ratio, rather than comparing one deviance with $n - p$; and use the Pearson statistic, whose expectation is $n - p$ times the dispersion whatever the size of the means, for the dispersion itself.

6 Exercises

Exercise 8.1. Prove that a sum of independent Poisson variables is Poisson, and explain why this makes an exposure natural for count data.

Solution. Proposition 8.2(iii): the generating function of the sum is the product $\prod_j e^{\mu_j(s-1)} = e^{(\sum \mu_j)(s-1)}$, which is the generating function of $\text{Pois}(\sum_j \mu_j)$; two laws on the non-negative integers with the same generating function agree, the coefficients of a convergent power series being determined by it.

A policy observed for m periods, each contributing an independent $\text{Pois}(\mu)$ count, therefore has a $\text{Pois}(m\mu)$ total: the mean is proportional to the amount of exposure, which is what $\ln \mu_i = \ln E_i + \mathbf{x}_i' \boldsymbol{\beta}$ encodes with the coefficient of $\ln E_i$ fixed at 1. ■

Exercise 8.2. A Poisson regression of claims on policy characteristics uses `offset(log(E))` and reports a coefficient 0.28 for a binary covariate. Interpret it. What changes if $\ln E$ is entered as an ordinary covariate?

Solution. By Proposition 8.5(ii) the claim rate per unit of exposure is multiplied by $e^{0.28} = 1.323$ when the covariate switches on, the exposure held fixed: a 32% higher frequency. Entering $\ln E$ as a covariate estimates a coefficient γ and fits $\mu_i = E_i^\gamma e^{\mathbf{x}_i' \boldsymbol{\beta}}$. The model no longer says that the expected count is proportional to exposure, so a fitted “rate” $\mu_i/E_i = E_i^{\gamma-1} e^{\mathbf{x}_i' \boldsymbol{\beta}}$ depends on the size of the policy, and rates from policies of different exposure are not comparable. §5.2 shows $\hat{\gamma} = 1.20$ on real data, and doubling the exposure then multiplies the fitted claims by 2.30 rather than by 2. ■

Exercise 8.3. Derive the Poisson score and information, and state what the score equations force about the fitted values.

Solution. Theorem 8.6: $\nabla \ell = \sum_i \mathbf{x}_i (y_i - \mu_i)$ and $\mathcal{I} = \sum_i \mu_i \mathbf{x}_i \mathbf{x}_i'$, the Hessian being $-\mathcal{I}$ identically because $\partial \mu_i / \partial \boldsymbol{\beta} = \mu_i \mathbf{x}_i$ carries no y_i . Corollary 8.7: at the optimum $\sum_i x_{ij} \hat{\mu}_i = \sum_i x_{ij} y_i$ for every covariate. With an intercept the fitted counts add to the observed total, and within each level of a categorical covariate the fitted and observed counts agree. That is why a Poisson fit with a saturated categorical design reproduces the cell counts exactly. ■

Exercise 8.4. A Poisson fit reports $\hat{\beta} = 0.42$ with standard error 0.05, and the Pearson statistic is 3,600 on 900 degrees of freedom. Give the quasi-Poisson standard error and the corrected z statistic, and say what happens to the point estimate.

Solution. $\hat{\phi} = 3600/900 = 4$, so by Theorem 8.14 the standard error is $\sqrt{4} \times 0.05 = 0.10$ and the statistic falls from $0.42/0.05 = 8.4$ to $0.42/0.10 = 4.2$. The point estimate is unchanged: the quasi-Poisson estimator solves the same score equations $\mathbf{X}'(\mathbf{Y} - \boldsymbol{\mu}) = \mathbf{0}$, which involve no variance assumption.

The conclusion is unaffected here, both statistics being large, but a Poisson z of 2.5 would become 1.25 and the sign of the conclusion would change. ■

Exercise 8.5. Show that a Poisson mean drawn from a gamma distribution yields a negative binomial, and give the resulting mean and variance. Why can the Poisson not be tested against it with a χ^2_1 likelihood ratio?

Solution. Theorem 8.17: integrating the Poisson mass function against the $\text{Gamma}(\theta, \theta/\mu)$ density leaves $\Gamma(j + \theta)/\{j!\Gamma(\theta)\}$ times $\{\theta/(\theta + \mu)\}^\theta$ and $\{\mu/(\theta + \mu)\}^j$, which is (10). Conditioning

gives $\mathbf{E}[y] = \mu$ and $\text{var}(y) = \mathbf{E}[\lambda] + \text{var}(\lambda) = \mu + \mu^2/\theta$.

The Poisson is recovered only as $\theta \rightarrow \infty$ (Corollary 8.18), so it lies on the boundary of the negative binomial parameter space rather than in its interior. The likelihood ratio theorem of the prerequisites expands the score in a neighbourhood of the null value, which does not exist here, and its χ^2 conclusion fails. An information criterion, or the moment test of Proposition 8.11, avoids the difficulty. ■

Exercise 8.6. A zero-inflated Poisson has $\pi = 0.3$ and $\mu = 2$. Compute the mean, the variance and the probability of a zero, and verify that the variance exceeds the mean.

Solution. By Theorem 8.21(i), $\mathbf{E}[y] = (1 - 0.3) \times 2 = 1.4$ and

$$\text{var}(y) = 1.4 + 0.3 \times 0.7 \times 4 = 1.4 + 0.84 = 2.24,$$

which exceeds the mean by $\pi(1 - \pi)\mu^2 = 0.84$, positive for any $\pi \in (0, 1)$.

The probability of a zero is $\pi + (1 - \pi)e^{-\mu} = 0.3 + 0.7e^{-2} = 0.3 + 0.0947 = 0.3947$, against $e^{-2} = 0.135$ for a Poisson with the same μ and 0.2466 for a Poisson with the same **mean** 1.4. ■

Exercise 8.7. Show that the hurdle log-likelihood separates into two pieces, and deduce that with an intercept-only zero part the fitted number of zeros equals the observed number.

Solution. Theorem 8.22: substituting (13) and using $\ln k_i = \ln(1 - \pi_i) - \ln(1 - g_i(0))$ collects the terms into a Bernoulli log-likelihood for $\mathbb{1}\{y_i = 0\}$ and a zero-truncated count log-likelihood, with no parameter in both.

With only an intercept in the first piece the Bernoulli maximum likelihood estimate is the sample proportion of zeros, so $n\hat{\pi}$ is the observed number of zeros (Corollary 8.23). The zero-inflated model has no such property: its zero probability $\pi_i + (1 - \pi_i)e^{-\mu_i}$ draws on both parts, which are fitted jointly. ■

Exercise 8.8. For the linear exponential family, prove $\mathbb{E}[y] = b'(\theta)$ and $\text{var}(y) = \phi b''(\theta)$ and identify the variance function of the gamma.

Solution. Theorem 8.25: the bound $|y|^k e^{y\theta/\phi} \leq k!(\phi/\delta)^k \{e^{y(\theta_0+2\delta)/\phi} + e^{y(\theta_0-2\delta)/\phi}\}$ dominates the first two θ -derivatives of the integrand uniformly near θ_0 , so the differentiation-under-the-integral theorem of the prerequisites applies twice to $A(\theta) = \int e^{y\theta/\phi} d\nu$; then $b' = \phi A'/A = \mathbb{E}[y]$ and

$$b'' = \phi(A''A - (A')^2)/A^2 = \text{var}(y)/\phi.$$

For the gamma, $b(\theta) = -\ln(-\theta)$ on $\theta < 0$, so $b'(\theta) = -1/\theta = \mu$ and $b''(\theta) = 1/\theta^2 = \mu^2$: the variance function is $V(\mu) = \mu^2$, and $\text{var}(y) = \phi\mu^2$. The coefficient of variation $\sqrt{\text{var } y}/\mathbf{E} y = \sqrt{\phi}$ is constant, so a gamma with a logarithmic link is the standard model for a severity. ■

Exercise 8.9. Write down the IRLS weights and working response for a general link, and check that they reduce to the Poisson case of §1.3 and to the logistic case of Unit 7.

Solution. Theorem 8.30: $W_i = w_i(d\mu_i/d\eta_i)^2/\{\phi V(\mu_i)\}$ and $z_i = \eta_i + (y_i - \mu_i)/(d\mu_i/d\eta_i)$.
 Poisson with a log link: $\phi = 1$, $w_i = 1$, $V(\mu) = \mu$ and $d\mu/d\eta = \mu$, so $W_i = \mu_i^2/\mu_i = \mu_i$ and $z_i = \eta_i + (y_i - \mu_i)/\mu_i$, which is Theorem 8.8.
 Logistic: $\phi = 1$, $V(\pi) = \pi(1 - \pi)$ and $d\pi/d\eta = \pi(1 - \pi)$, so $W_i = \pi_i(1 - \pi_i)$ and $z_i = \eta_i + (y_i - \pi_i)/\{\pi_i(1 - \pi_i)\}$, which is the weighted least squares step of Unit 7. In both the link is canonical, so $d\mu/d\eta$ equals $V(\mu)$ and the weight collapses to $V(\mu_i)$. ■

Exercise 8.10. Derive the Poisson deviance from the general definition and explain why the deviance of a correct model need not be close to $n - p$.

Solution. Proposition 8.32: with $b(\theta) = e^\theta$ and $\theta = \ln \mu$, the saturated fit solves $e^{\tilde{\theta}_i} = y_i$, so $d_i = 2\{y_i \ln y_i - y_i \ln \hat{\mu}_i - y_i + \hat{\mu}_i\}$ and summing gives (26).

The reference $n - p$ comes from an approximation that improves as each μ_i grows, not as n grows: the deviance is a sum of n terms, each a discrepancy for one observation, and with μ_i of order one those terms are not close to squared standard normals. §5.4 shows the mean deviance of a correct intercept-only Poisson fit at 1,683 against $n - p = 1,999$ when $\mu = 0.3$, and at 2,003 when $\mu = 50$.



Exercise 8.11. Show that the compound Poisson–gamma sum with the parametrisation (30) has mean μ , variance $\phi\mu^p$ and an atom at zero, and say what happens as $p \rightarrow 1$ and $p \rightarrow 2$.

Solution. Theorem 8.37: conditioning on N gives $E[S] = \lambda\alpha/\gamma$ and $\text{var}(S) = \lambda\alpha(1 + \alpha)/\gamma^2$; substituting $\alpha/\gamma = \phi(2 - p)\mu^{p-1}$ and $1 + \alpha = 1/(p - 1)$ turns these into μ and $\phi\mu^p$. The atom is $P\{N = 0\} = e^{-\lambda}$ with $\lambda = \mu^{2-p}/\{\phi(2 - p)\}$, since every gamma summand is strictly positive. As $p \downarrow 1$, $\alpha = (2 - p)/(p - 1) \rightarrow \infty$ and the gamma summand degenerates to a constant while $\lambda \rightarrow \mu/\phi$: the limit is a scaled Poisson, and $V(\mu) = \mu$. As $p \uparrow 2$, $\lambda \rightarrow \infty$ so the atom at zero

vanishes, $\alpha \rightarrow 0$, and the limit is the gamma with $V(\mu) = \mu^2$. The index p therefore interpolates between a pure count and a pure positive severity, and $1 < p < 2$ is the range with both an atom and a continuous part. ■

Exercise 8.12. In R: (a) reproduce a Poisson `glm` by iteratively reweighted least squares; (b) show that the quasi-Poisson coefficients equal the Poisson ones and its standard errors are $\sqrt{\hat{\phi}}$ times as large; (c) simulate a Poisson mixed over a gamma and match the negative binomial mass function.

Solution. (a) is §4.1, (b) §4.3 and (c) §4.4. In (b) the two statements are exact rather than approximate, and the printed differences are at the level of floating-point arithmetic; the interesting number is the spread of the sandwich ratios, from 0.16 to 7.07, which says that the single constant $\hat{\phi}$ of quasi-Poisson does not describe this variance function either. ■